Mathematics for Data Science

Lecture 5 - Normed spaces

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Previously covered topics

- (Lecture 1) Vector spaces, subspaces, linear transformations. Rank, image, kernel
- (Lecture 2) Matrices, link with linear transformations, linear systems
- Range, rank, kernel of a matrix, inverse of a matrix
- (Lecture 3) Determinant
- (Lecture 4) Eigendecomposition, Diagonalization.

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In this lecture: Normed spaces, metric spaces, inner product spaces. Orthogonal complement, orthogonal matrix (again!).
Spectral theorem ; positive (semi-)definite matrices, Gram matrix ; Singular value decomposition.

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Inner product

Definition

An **inner product** on a real vector space V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying, for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and all $\alpha, \beta \in \mathbb{R}$:

(i) (symmetric) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$

(ii) (bilinear) $\langle \alpha \mathbf{x} + \beta \mathbf{y}, \mathbf{z} \rangle = \alpha \langle \mathbf{x}, \mathbf{z} \rangle + \beta \langle \mathbf{y}, \mathbf{z} \rangle$

(iii) (positive-definite) $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$, with equality if and only if $\mathbf{x} = 0_V$

Vocabulary: A vector space endowed with an inner product is called an **inner product space**, or a **pre-Hilbert space** (FR: *espaces préhilbertiens*).

NB: in the above definition we have stated linearity in the first slot; with symmetry this implies linearity in the second.

Scalar product (dot product)

The usual **scalar product** $(.|.)$ defined on $\mathbb{R}^n$ is an inner product.

$$(.|.) : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}, (x_1, x_2, \dots x_n), (y_1, y_2, \dots y_n) \mapsto \sum_{i=1}^{n} x_i y_i$$

The inner product on $\mathbb{R}^n$ is also often written $\mathbf{x} \cdot \mathbf{y}$ (hence the alternate name **dot product**).

NB: If $x, y \in \mathbb{R}^n$, the dot product can be expressed as:

$$\langle x, y \rangle = x^\top y.$$

i.e. this inner product is a special case of matrix multiplication where we regard the resulting 1×1 matrix as a scalar.

NB: for $\alpha_1, \dots \alpha_n \in \mathbb{R}_{>0}$ (strictly positive), the map defined on $\mathbb{R}^n$ as follows is also an inner product.

$$(x_1, x_2, \dots x_n), (y_1, y_2, \dots y_n) \mapsto \sum_{i=1}^{n} \alpha_i x_i y_i$$

Properties of the Dot Product ($\mathbb{R}^n$)

Using the previous definition of the scalar/dot product in $\mathbb{R}^n$, prove the following properties.

Properties of the Dot Product

For all $x, y, z \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$:

1. Symmetry: $\langle x, y \rangle = \langle y, x \rangle$.
2. Homogeneity: $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$.
3. Linearity in the first argument: $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
4. Linearity in the second argument: $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$.

Proof: exercise.

Another example of inner product

Example with functions

Another example of inner product

Example with functions

Example

Take the vector space of continuous real-valued functions defined over a segment $[a, b]$ of $\mathbb{R}$, with $a < b$, $E = \mathcal{C}^0([a, b], \mathbb{R})$. Prove that the map $(.|.)$ defined on E^2 as follows is an inner product .

$$(f|g) = \int_a^b f(x)g(x)\,dx$$

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$$(f|g) = \int_a^b f(x)g(x)\,dx$$

NB: Symmetry, bilinearity, positive-definite.
f^2 is continuous too on $[a, b]$. If $f \in \mathcal{C}^0([a, b])$ and $\int_a^b f(x)^2\,dx = 0$, then by continuity f is the null function on $[a, b]$.

Another example of inner product

Example with matrices

Another example of inner product

Example with matrices

Example

Consider the vector space of real square matrices of size n, $\mathcal{M}_n(\mathbb{R})$. We define the following application from $\mathcal{M}_n(\mathbb{R}) \times \mathcal{M}_n(\mathbb{R})$ to $\mathbb{R}$:
$(A, B) \mapsto \text{tr}\left(A^\top B\right)$.
Prove that this application is an inner product on $\mathcal{M}_n(\mathbb{R})$.

Another example of inner product

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Example

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$(A, B) \mapsto \text{tr}\left(A^\top B\right)$.
Prove that this application is an inner product on $\mathcal{M}_n(\mathbb{R})$.
NB: again prove symmetry, bilinearity, positive-definiteness.

NB: positive-definiteness follows from

$$\text{tr}\left(A^\top A\right) = \sum_{i,j} a_{ij}^2.$$

Orthogonal vectors

Definition

Two vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be **orthogonal** if their inner product is zero

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0$$

Notation : we can write $\mathbf{x} \perp \mathbf{y}$ for short.

NB: **Orthogonality** generalizes the notion of **perpendicularity** we use in 2D.

Writing the dot product using matrix notation (again)

Let $x, y \in \mathbb{R}^n$. The dot product can be expressed as:

$$\langle x, y \rangle = x^\top y.$$

This is a special case of matrix multiplication where we regard the resulting 1×1 matrix as a scalar. Example: if $x = (x_1, x_2, x_3)^\top$ and $y = (y_1, y_2, y_3)^\top$, then

$$\langle x, y \rangle = x^\top y = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = x_1 y_1 + x_2 y_2 + x_3 y_3.$$

Example

Let $A \in \mathcal{M}_n(\mathbb{R})$ a square matrix and $x, y \in \mathbb{R}^n$. Show that $\langle Ax, y \rangle = \langle x, A^\top y \rangle$.

Writing the dot product using matrix notation (again)

Let $x, y \in \mathbb{R}^n$. The dot product can be expressed as:

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Example

Let $A \in \mathcal{M}_n(\mathbb{R})$ a square matrix and $x, y \in \mathbb{R}^n$. Show that $\langle Ax, y \rangle = \langle x, A^\top y \rangle$. $\langle Ax, y \rangle = (Ax)^\top y = x^\top A^\top y = \langle x, A^\top y \rangle$.

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Metric spaces

A **metric space** is a set together with a notion of **distance** between its elements, usually called **points**. The distance is measured by a function called a **metric** or **distance** function.

Definition (Metric)

A **metric** on a set S is a function $d : S \times S \to \mathbb{R}^+$ that satisfies, for all $x, y, z \in S$:

(i) (Positivity) $d(x, y) \geq 0$, with equality if and only if $x = y$

(ii) (Symmetry) $d(x, y) = d(y, x)$

(iii) (Triangle inequality) $d(x, z) \leq d(x, y) + d(y, z)$

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(iii) (Triangle inequality) $d(x, z) \leq d(x, y) + d(y, z)$

Examples of metric spaces ?

Metric spaces - Examples

Example

- The real numbers with the distance function $d(x, y) = |y - x|$ given by the absolute difference between two real numbers
- For a given $n > 0$, the **Hamming distance** is a metric on the set of the words of length n, e.g. the set of 100-character Unicode strings can be equipped with the Hamming distance, which measures the number of characters that need to be changed to get from one string to another.

NB: The Hamming distance actually comes from information theory, where it is used to count the minimum number of errors that could have transformed one string into the other.

Metric spaces - Examples

Example

The Euclidean plane $\mathbb{R}^2$ can be equipped with many different metrics:

- The usual Euclidean distance from high school :
  $d_2((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- The Manhattan (or "taxi-cab") distance :
  $d_1((x_1, y_1), (x_2, y_2)) = |x_2 - x_1| + |y_2 - y_1|$
- The maximum distance :
  $d_\infty((x_1, y_1), (x_2, y_2)) = \max\{|x_2 - x_1|, |y_2 - y_1|\}$
- A discrete metric like the following : $d(p, q) = \begin{cases} 0, & \text{if } p = q, \\ 1, & \text{otherwise.} \end{cases}$

NB: More generally, we call **Euclidean space** a vector space defined over $\mathbb{R}$, that has a finite dimension and is endowed with an inner product.

ML illustration: metrics and nearest neighbors

A metric gives a notion of distance between data points.
Given a query point x_q and a dataset

$$x^{(1)}, \ldots, x^{(m)},$$

a nearest-neighbor classifier predicts using the closest data points:

$$i^\star = \arg\min_i d(x_q, x^{(i)}).$$

The choice of metric matters:

- Euclidean distance compares geometric proximity;
- Manhattan distance may be better for sparse or grid-like features;
- Hamming distance compares binary strings or discrete codes.

Interpretation: changing the metric changes what the algorithm considers "similar".

Remarks

- Two different metrics will give us two different "measures" of distances.
- Not all metric spaces are vector spaces !
- A key motivation for metrics is that they allow limits to be defined for mathematical objects other than real numbers.
  e.g. we say that a sequence $\{x_n\} \subseteq S$ converges to the limit x if for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $d(x_n, x) < \epsilon$ for all $n \geq N$.

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Norms

Some (but not all !) metric spaces are **normed spaces**. They are defined on vector spaces and endowed with a **norm** function.

Definition (Norm)

A **norm** on a real vector space V is a function $\|\cdot\| : V \to \mathbb{R}^+$ that satisfies

(i) $\|\mathbf{x}\| \geq 0$, with equality if and only if $\mathbf{x} = 0_V$

(ii) $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$

(iii) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ (the **triangle inequality** again)

for all $\mathbf{x}, \mathbf{y} \in V$ and all $\alpha \in \mathbb{R}$.

A vector space endowed with a norm is called a **normed vector space**, or simply a **normed space**.

Norm induced by an inner product

Remark

Any inner product on V induces a norm on V:

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$$

NB 1: the axioms for norms are satisfied under this definition and follow from the axioms for inner products. Therefore *any inner product space is also a normed space (and hence also a metric space).*
NB 2: positivity and homogeneity follow directly from the inner-product axioms. The triangle inequality follows from the Cauchy–Schwarz inequality, proved after.

Norms are useful to measure distances :

$$d(x, y) = \|\mathbf{x} - \mathbf{y}\| = \|\mathbf{y} - \mathbf{x}\|$$

Two different norms will give us two different "measures" of distances.

Norm induced by the dot product on $\mathbb{R}^n$

Example

Verify that the two-norm $\|\cdot\|_2$ (Euclidean norm) on $\mathbb{R}^2$ is induced by the dot product.

$$\langle x, x \rangle = \|x\|_2^2.$$

NB: "two-norm", l_2, L_2

Examples of norms on $\mathbb{R}^n$

$$L_1 : \|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i|$$

$$L_2 : \|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2} \quad \text{(Euclidean norm)}$$

$$L_p : \|\mathbf{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} \quad (p \geq 1)$$

$$L_\infty : \|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|$$

NB: The 1- and 2-norms are special cases of the p-norm, and the ∞-norm is the limit of the p-norm as p tends to infinity.
We require $p \geq 1$ because the triangle inequality fails to hold for $0 < p < 1$.

Geometric interpretation of norms in $\mathbb{R}^2$

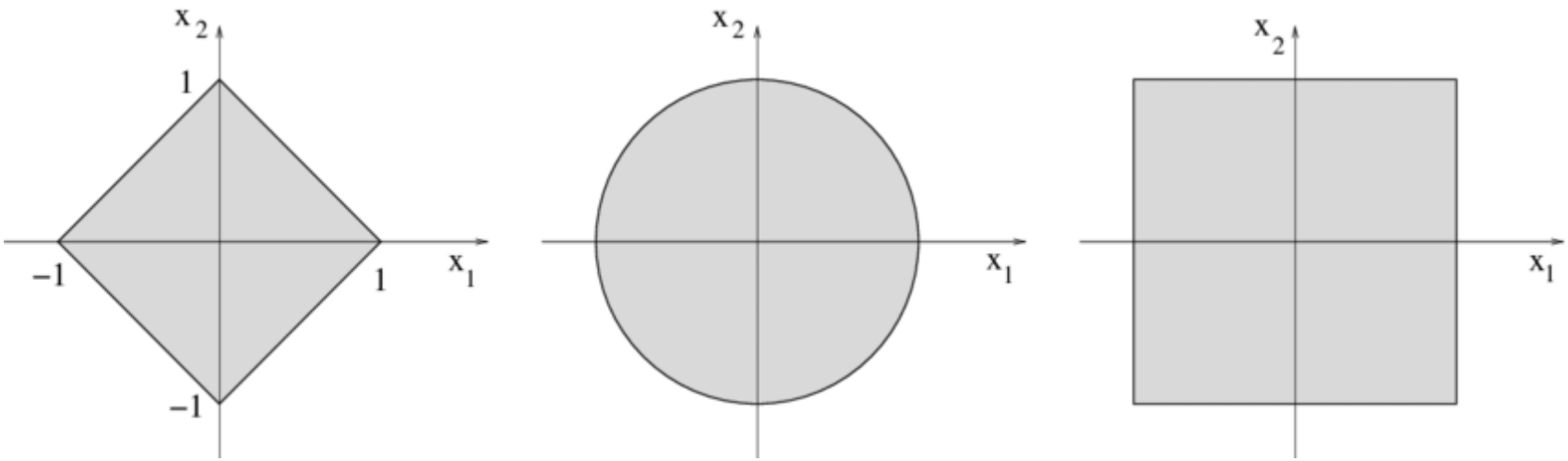


Figure 1: Unit balls for different norms in $\mathbb{R}^2$

For each subfigure, give the corresponding norm.

Geometric interpretation of norms in $\mathbb{R}^2$

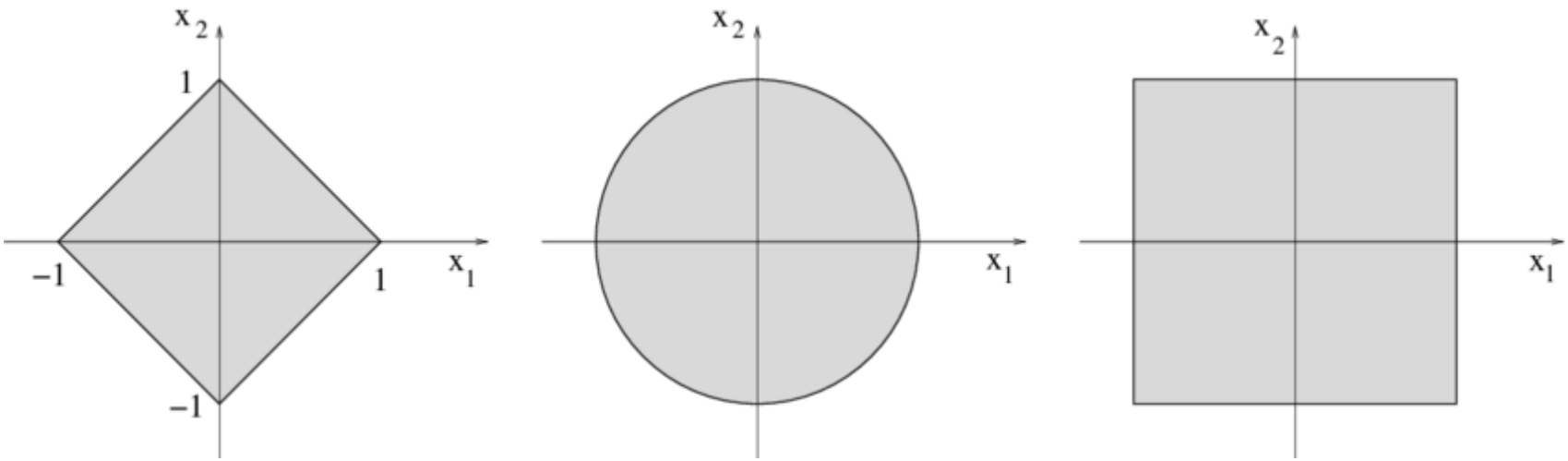


Figure 1: Unit balls for different norms in $\mathbb{R}^2$

For each subfigure, give the corresponding norm. The unit ball for the 2-norm is a circle, for the 1-norm a square rotated by 45 degrees, and for the infinity norm a square aligned with the axes.

Geometric interpretation of norms in $\mathbb{R}^2$

For the angle θ between x and y (vectors different from the null vector):

$$\langle x, y \rangle = \|x\|_2 \, \|y\|_2 \cos\theta.$$

Reminder : If $\langle x, y \rangle = 0$, the vectors x and y are said to be *orthogonal*.

Example

Let $v_1, v_2, \ldots v_p \in \mathbb{R}^n$ be p nonzero orthogonal vectors. Show that they are linearly independent.

Geometric interpretation of norms in $\mathbb{R}^2$

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$$\langle x, y \rangle = \|x\|_2 \, \|y\|_2 \cos\theta.$$

Reminder : If $\langle x, y \rangle = 0$, the vectors x and y are said to be *orthogonal*.

Example

Let $v_1, v_2, \ldots v_p \in \mathbb{R}^n$ be p nonzero orthogonal vectors. Show that they are linearly independent. Consider a linear combination equal to zero: $\sum_{i=1}^{p} \alpha_i v_i = 0$. Taking the inner product with $v_j, 1 \leq j \leq p$ gives $\alpha_j \|v_j\|^2 = 0$, hence $\alpha_j = 0$ for all j. So the vectors are linearly independent (and $p \leq n$).

ML illustration: norms as regularizers

Norms are widely used to regularize ML models.
For a parameter vector $w \in \mathbb{R}^d$, one often minimizes

$$\text{loss}(w) + \lambda \|w\|.$$

Different norms have different effects:

- $\|w\|_2^2$ penalizes large weights smoothly;
- $\|w\|_1$ encourages sparse solutions;
- $\|w\|_\infty$ controls the largest coordinate.

Interpretation: a norm encodes what it means for a parameter vector to be "small".

Pythagorean Theorem

The Pythagorean theorem generalizes to arbitrary inner product spaces.

Theorem (Pythagorean Theorem)

Let V be an inner-product vector space, and $x, y \in V$. If $\mathbf{x} \perp \mathbf{y}$, then

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$$

Proof: exercise.
Suppose $\mathbf{x} \perp \mathbf{y}$, i.e. $\langle \mathbf{x}, \mathbf{y} \rangle = 0$...

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Proof: exercise.
Suppose $\mathbf{x} \perp \mathbf{y}$, i.e. $\langle \mathbf{x}, \mathbf{y} \rangle = 0$... It follows:

$$\|\mathbf{x} + \mathbf{y}\|^2 = \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle$$

Then, distributing the product and identifying the norms:

$$\langle \mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle + \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$$

Cauchy-Schwarz inequality

Proposition

Let V be an inner product space. For all $\mathbf{x}, \mathbf{y} \in V$,

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$$

NB: this equality holds exactly iff $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent (i.e. are scalar multiples of each other, including the case when at least one of them is null).

Proof: exercise.

ML illustration: Cauchy–Schwarz and cosine similarity

Cosine similarity can be expressed in terms of the inner product and the 2-norm:

$$\text{cosine-similarity}(x, y) = \frac{\langle x, y \rangle}{\|x\|_2 \|y\|_2}.$$

Since the Cauchy–Schwarz inequality says

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2.$$

We see that the cosine similarity is always between -1 and 1:

$$-1 \leq \frac{x^\top y}{\|x\|_2 \|y\|_2} \leq 1.$$

ML interpretation: this makes cosine similarity a practical measure of alignment between embeddings.

It is widely used in retrieval and self-supervised learning.

Cauchy-Schwarz inequality (exercise)

If $\mathbf{y} = 0$, the result is trivial. Assume $\mathbf{y} \neq 0$ and consider

$$\mathbf{z} = \mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2} \mathbf{y}.$$

Since $\langle \mathbf{z}, \mathbf{z} \rangle \geq 0$, we get

$$0 \leq \left\| \mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2} \mathbf{y} \right\|^2 = \|\mathbf{x}\|^2 - \frac{\langle \mathbf{x}, \mathbf{y} \rangle^2}{\|\mathbf{y}\|^2}.$$

Therefore

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \|\mathbf{x}\|^2 \|\mathbf{y}\|^2.$$

Taking square roots gives

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$

Equivalence of norms in a finite-dimensional vector space (bonus)

For any given finite-dimensional vector space V, all norms on V are equivalent in the sense that for all $\mathbf{x} \in V$, for two norms $\|\cdot\|_A$ and $\|\cdot\|_B$, there exist constants $\alpha, \beta > 0$ such that

$$\alpha\|\mathbf{x}\|_A \leq \|\mathbf{x}\|_B \leq \beta\|\mathbf{x}\|_A$$

NB: the constants depend on the norms, not on the vector. **Therefore convergence in one norm implies convergence in any other norm.**

This is not a general property e.g. may not apply in infinite-dimensional vector spaces such as function spaces!

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"Entry-wise" matrix norms (bonus)

"Entry-wise" matrix norms treat an $m \times n$ matrix as a vector of size $m \cdot n$, and use one of the familiar vector norms we have seen before. For example, using the L_p-norm for vectors:

$$\|A\|_{p,p} = \|\mathrm{vec}(A)\|_p = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^p \right)^{1/p}$$

This is a different norm from the induced p-norm (see after)

Example

The special case $p = 2$ is the **Frobenius norm**.

$$\|A\|_{\mathrm{F}} = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2}$$

ML illustration: Frobenius norm as reconstruction loss

Let $X \in \mathbb{R}^{m \times d}$ be a data matrix, and let $\widehat{X}$ be a reconstruction.
A common reconstruction error is

$$\|X - \widehat{X}\|_F^2 = \sum_{i=1}^{m} \sum_{j=1}^{d} (X_{ij} - \widehat{X}_{ij})^2.$$

Interpretation: the Frobenius norm measures the total squared reconstruction error across all entries.

It is used in matrix factorization, PCA, autoencoders, and denoising.

Matrix norms induced by vector norms (bonus)

Definition (Matrix norm induced by vector norms)

Let $\|\cdot\|_\alpha$ be a norm on $\mathbb{R}^n$ and $\|\cdot\|_\beta$ a norm on $\mathbb{R}^m$. For $A \in \mathbb{R}^{m\times n}$, define

$$\|A\|_{\alpha,\beta} = \sup_{x\neq 0} \frac{\|Ax\|_\beta}{\|x\|_\alpha} = \sup_{\|x\|_\alpha \leq 1} \|Ax\|_\beta.$$

NB 1: This norm can be interpreted as measuring how much the mapping induced by A "stretches" vectors. (Useful for stability analysis of linear systems! this is called the **gain** of the system.)
NB 2: Given a vector norm $\|\cdot\|$ on $\mathbb{R}^n$, a *matrix norm* can be defined by:

$$\|A\| = \sup_{x\neq 0} \frac{\|Ax\|}{\|x\|}.$$

Matrix norms induced by vector norms (bonus)

If the L_p-norm for vectors ($1 \leq p$) is used for both spaces $\mathbb{R}^n$ and $\mathbb{R}^m$ then the corresponding induced norm is:

$$\|A\|_p = \sup\{\|Ax\|_p : x \in \mathbb{R}^n \text{ such that } \|x\|_p \leq 1\}$$

Example

- Case $p = 1$: the maximum absolute *column sum* of the matrix

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^{m} |a_{ij}|$$

- Case $p = \infty$: the maximum absolute *row sum* of the matrix.

$$\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^{n} |a_{ij}|$$

Matrix norms induced by vector norms (bonus)

Example

- Case $p = 2$: the **spectral norm**. Not to be confounded with the **Frobenius norm** !

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^\top A)} = \sigma_{\max}(A).$$

$$\|A\|_2 = \sigma_{\max}(A) \leq \|A\|_{\mathrm{F}}$$

NB: In the complex case, one uses the conjugate transpose A^*. $\sigma_{\max}(A)$ the highest singular value of A (see section on SVD).

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Orthogonal basis

Definition

A set of vectors $\{e_1, \ldots, e_n\}$ from a finite-dimensional inner-product space E is an *orthogonal basis* of E if it is a basis and $\langle e_i, e_j \rangle = 0$ whenever $i \neq j$.

Theorem

Every finite-dimensional inner product space admits an orthonormal basis.

NB: *orthogonal* vs *orthonormal*.

Orthogonal complement

Definition

Let F be a subspace of a vector space E endowed with an inner product. The *orthogonal complement* of F, denoted $F^{\perp}$, is:

$$F^{\perp} = \{x \in E \mid \langle x, v \rangle = 0 \text{ for all } v \in F\}.$$

Example

In $\mathbb{R}^3$, what is the orthogonal complement of the subspace spanned by the vector $(1, 1, 0)$?

Orthogonal complement

Definition

Let F be a subspace of a vector space E endowed with an inner product. The *orthogonal complement* of F, denoted $F^{\perp}$, is:

$$F^{\perp} = \{x \in E \mid \langle x, v \rangle = 0 \text{ for all } v \in F\}.$$

Example

In $\mathbb{R}^3$, what is the orthogonal complement of the subspace spanned by the vector $(1, 1, 0)$? It is the plane defined by the equation $x + y = 0$, i.e. the set of vectors (x, y, z) such that $x + y = 0$. So $\text{span}((1, 1, 0))s^{\perp} = \text{span}((1, -1, 0), (0, 0, 1))$.

Orthogonal complement

Proposition

Let F be a subspace of a finite-dimensional inner product space E. Then:

(i) $F^\perp$ is a subspace of E.

(ii) $F \cap F^\perp = \{0_E\}$.

(iii) $\dim(F) + \dim(F^\perp) = \dim(E)$.

(iv) $(F^\perp)^\perp = F$.

Proof: exercise session.

Link with linear maps: Projectors

Definition (Projector)

A linear map $p \in \mathcal{L}(E)$ is called a **projector** if $p \circ p = p$.

Definition (Orthogonal projection)

Let E be a euclidean vector space, and F a subspace of E. We call **orthogonal projection** onto F the linear map $p_F : E \to E$ such that for all $\mathbf{x} \in E$, $p_F(\mathbf{x}) \in F$ and

$$\mathbf{x} - p_F(\mathbf{x}) \in F^{\perp}$$

NB: $p_F(\mathbf{x})$ is called the **projection** of $\mathbf{x}$ onto F.
You can check that for all $\mathbf{x} \in E$,

$$p_F(p_F(\mathbf{x})) = p_F(\mathbf{x})$$

Properties of projectors

Orthogonal projections are contractions

For all $\mathbf{x} \in E$,

$$\|p_F(\mathbf{x})\| \leq \|\mathbf{x}\|$$

with equality if and only if $\mathbf{x} \in F$.

Proof: exercise.

Properties of projectors

Orthogonal projections are contractions

For all $\mathbf{x} \in E$,

$$\|p_F(\mathbf{x})\| \leq \|\mathbf{x}\|$$

with equality if and only if $\mathbf{x} \in F$.

Proof: exercise. Let $x \in E$. We can write

$$x = p_F(x) + (x - p_F(x))$$

with $p_F(x) \in F$ and $x - p_F(x) \in F^{\perp}$. By the Pythagorean theorem, we have

$$\|x\|^2 = \|p_F(x)\|^2 + \|x - p_F(x)\|^2 \geq \|p_F(x)\|^2$$

with equality if and only if $x - p_F(x) = 0$, i.e. $x \in F$.

Properties of projectors

Proposition

Let $p \in \mathcal{L}(E)$ a projector (i.e. $p \circ p = p$). The following are equivalent:

(i) p is an orthogonal projector onto $F = \text{range}(p)$ and $\text{Ker}(p) = F^{\perp}$

(ii) For all $\mathbf{x}, \mathbf{y} \in E$, $\langle p(\mathbf{x}), y - p(y) \rangle = 0$

(iii) For all $\mathbf{x}, \mathbf{y} \in E$, $\langle p(\mathbf{x}), y \rangle = \langle \mathbf{x}, p(y) \rangle$

Proof: exercise.

Projection onto a subspace (via an orthonormal basis)

Proposition

Let E be a euclidean vector space, F a subspace of E and $\{e_1, \ldots, e_k\}$ an orthonormal basis of F. For all $\mathbf{x} \in E$, the orthogonal projection of $\mathbf{x}$ onto F is given by:

$$p_F(\mathbf{x}) = \sum_{i=1}^{k} \langle \mathbf{x}, e_i \rangle e_i$$

Distance to a subspace

Definition

Let E be a euclidean vector space, and F a subspace of E. The **distance** from a vector $\mathbf{x} \in E$ to the subspace F is defined as:

$$d(\mathbf{x}, F) = \inf_{\mathbf{y} \in F} \|\mathbf{x} - \mathbf{y}\|$$

Proposition

Let E be a euclidean vector space, F a subspace of E and p_F the orthogonal projection onto F. Then for all $\mathbf{x} \in E$, $p_F(\mathbf{x})$ is the unique vector in F satisfying:

$$d(\mathbf{x}, F) = \|\mathbf{x} - p_F(\mathbf{x})\|$$

.

Orthogonal endomorphism

Definition

An endomorphism $f \in \mathcal{L}(E)$ is said to be **orthogonal** if it preserves the inner product, i.e. for all $\mathbf{x}, \mathbf{y} \in E$,

$$\langle f(\mathbf{x}), f(\mathbf{y}) \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$$

Proposition

An endomorphism $f \in \mathcal{L}(E)$ is orthogonal if and only if it preserves the norm, i.e. for all $\mathbf{x} \in E$,

$$\|f(\mathbf{x})\| = \|\mathbf{x}\|$$

Proof: exercise.

Orthogonal endomorphism

Proof: (orthogonal $\Longrightarrow$ norm-preserving):

$$\|f(\mathbf{x})\|^2 = \langle f(\mathbf{x}), f(\mathbf{x})\rangle = \langle \mathbf{x}, \mathbf{x}\rangle = \|\mathbf{x}\|^2$$

Orthogonal endomorphism

Proof: (orthogonal $\implies$ norm-preserving):

$$\|f(\mathbf{x})\|^2 = \langle f(\mathbf{x}), f(\mathbf{x})\rangle = \langle \mathbf{x}, \mathbf{x}\rangle = \|\mathbf{x}\|^2$$

For the converse, (norm-preserving $\implies$ orthogonal): we use that f is linear and the polarization identity.
Remark: little trick

$$4\langle \mathbf{x}, \mathbf{y}\rangle = \|\mathbf{x}+\mathbf{y}\|^2 - \|\mathbf{x}-\mathbf{y}\|^2 = \|f(\mathbf{x}+\mathbf{y})\|^2 - \|f(\mathbf{x}-\mathbf{y})\|^2$$

$$= \|f(\mathbf{x})+f(\mathbf{y})\|^2 - \|f(\mathbf{x})-f(\mathbf{y})\|^2 = 4\langle f(\mathbf{x}), f(\mathbf{y})\rangle$$

Link with orthogonal matrices

Definition

Let $Q \in \mathcal{M}_n(\mathbb{R})$. Q is called an orthogonal matrix if the associated linear map $f_Q : \mathbb{R}^n \to \mathbb{R}^n$ defined by $f_Q(\mathbf{x}) = Q\mathbf{x}$ is an orthogonal endomorphism.

The columns of an orthogonal matrix Q are pairwise orthonormal. They form an orthonormal basis of $\mathbb{R}^n$. This definition also implies that

$$Q^\top Q = QQ^\top = I_n$$

or equivalently, $Q^\top = Q^{-1}$.
Orthogonal matrices preserve inner products:

$$(Q\mathbf{x})^\top (Q\mathbf{y}) = \mathbf{x}^\top Q^\top Q\mathbf{y} = \mathbf{x}^\top \mathbf{I}_n \mathbf{y} = \mathbf{x}^\top \mathbf{y}$$

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Symmetric endomorphisms and matrices

Definition

Let E be a euclidean vector space. An endomorphism $f \in \mathcal{L}(E)$ is said to be **symmetric** if for all $\mathbf{x}, \mathbf{y} \in E$,

$$\langle f(\mathbf{x}), \mathbf{y} \rangle = \langle \mathbf{x}, f(\mathbf{y}) \rangle$$

NB: If $E = \mathbb{R}^n$ with the standard inner product, then f is symmetric if and only if its matrix in the standard basis is a symmetric matrix, i.e. $A^\top = A$.

Example

We have seen that the orthogonal projection onto a given subspace is a symmetric endomorphism.

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Spectral theorem

Theorem (Spectral Theorem for endomorphisms)

Let E be a finite-dimensional euclidean vector space, and $f \in \mathcal{L}(E)$ a symmetric endomorphism. Then there exists an orthonormal basis of E consisting of eigenvectors of f.

Theorem (Spectral Theorem for matrices)

If A is a symmetric $n \times n$ matrix, then A is **orthogonally diagonalizable**, *i.e. there exists an orthogonal matrix V and a diagonal matrix D such that:*

$$A = VDV^T,$$

where the diagonal entries of D are the n real eigenvalues of A.

Practical application: any symmetric matrix is diagonalizable in an orthonormal basis, a decomposition called **spectral decomposition**.

Spectral theorem

Proof: exercise.

Spectral theorem

Proof: exercise. Let E be a finite-dimensional euclidean vector space, and $f \in \mathcal{L}(E)$ a symmetric endomorphism. We want to prove that there exists an orthonormal basis of E consisting of eigenvectors of f.

The proof proceeds in three steps:

- First, we prove that a symmetric endomorphism admits at least one eigenvalue and one eigenvector:

$$\mathrm{span}(f)_{\mathbb{R}} \neq \emptyset.$$

.

- Then, we prove that if F is a stable subspace of E under f, then its orthogonal complement $F^{\perp}$ is also stable under f.
- Finally, we prove the theorem by induction on n.

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Rayleigh quotients

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a symmetric matrix.

Definition

The expression $\mathbf{x}^\top \mathbf{A}\mathbf{x}$ is called a **quadratic form**.
The following quantity is called a **Rayleigh quotient**:

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{A}\mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

NB: the Rayleigh quotient is defined for all $\mathbf{x} \neq 0$.

Proposition

(i) **Scale invariance**: for any vector $\mathbf{x} \neq 0$ and any scalar $\alpha \neq 0$, $R_{\mathbf{A}}(\alpha \mathbf{x}) = R_{\mathbf{A}}(\mathbf{x})$.

(ii) If $\mathbf{x}$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ, then $R_{\mathbf{A}}(\mathbf{x}) = \lambda$.

Rayleigh quotients

Proof: exercise.

Rayleigh quotients

Proof: exercise.
(i) Let $\mathbf{x} \neq 0$ and $\alpha \neq 0$. Then

$$R_{\mathbf{A}}(\alpha \mathbf{x}) = \frac{(\alpha \mathbf{x})^\top \mathbf{A}(\alpha \mathbf{x})}{(\alpha \mathbf{x})^\top (\alpha \mathbf{x})} = \frac{\alpha^2 \mathbf{x}^\top \mathbf{A} \mathbf{x}}{\alpha^2 \mathbf{x}^\top \mathbf{x}} = \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = R_{\mathbf{A}}(\mathbf{x})$$

Rayleigh quotients

Proof: exercise.

(i) Let $\mathbf{x} \neq 0$ and $\alpha \neq 0$. Then

$$R_{\mathbf{A}}(\alpha \mathbf{x}) = \frac{(\alpha \mathbf{x})^\top \mathbf{A}(\alpha \mathbf{x})}{(\alpha \mathbf{x})^\top (\alpha \mathbf{x})} = \frac{\alpha^2 \mathbf{x}^\top \mathbf{A} \mathbf{x}}{\alpha^2 \mathbf{x}^\top \mathbf{x}} = \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = R_{\mathbf{A}}(\mathbf{x})$$

(ii) Let $\mathbf{x}$ be an eigenvector of $\mathbf{A}$ with eigenvalue λ. Then

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{\mathbf{x}^\top (\lambda \mathbf{x})}{\mathbf{x}^\top \mathbf{x}} = \frac{\lambda \mathbf{x}^\top \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \lambda$$

Rayleigh quotients

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a symmetric matrix.

Proposition

For any $\mathbf{x}$ such that $\|\mathbf{x}\|_2 = 1$,

$$\lambda_{\min}(\mathbf{A}) \leq \mathbf{x}^\top \mathbf{A}\mathbf{x} \leq \lambda_{\max}(\mathbf{A})$$

with equality if and only if $\mathbf{x}$ is a corresponding eigenvector.

Rayleigh quotients

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a symmetric matrix.

Proposition

For any $\mathbf{x}$ such that $\|\mathbf{x}\|_2 = 1$,

$$\lambda_{\min}(\mathbf{A}) \leq \mathbf{x}^\top \mathbf{A}\mathbf{x} \leq \lambda_{\max}(\mathbf{A})$$

with equality if and only if $\mathbf{x}$ is a corresponding eigenvector.

Theorem (Rayleigh quotient bound)

For all $\mathbf{x} \neq 0$,

$$\lambda_{\min}(\mathbf{A}) \leq R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_{\max}(\mathbf{A})$$

with equality if and only if $\mathbf{x}$ *belongs to the eigenspace associated with* $\lambda_{\min}(A)$ *or* $\lambda_{\max}(A)$, *respectively.*

Proof: (sketch). Since $\mathbf{x}^\top \mathbf{A}\mathbf{x} = R_{\mathbf{A}}(\mathbf{x})$ for unit $\mathbf{x}$, using the above proposition, thebound follows.

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Positive (semi-)definite matrices

Definition (Positive (semi-)definite matrix)

An $n \times n$ matrix A is *positive definite* if:

1. A is symmetric ($A^T = A$),
2. For all $x \in \mathbb{R}^n \setminus \{0\}$, $x^T A x > 0$.

It is *positive semi-definite*

1. if A is symmetric,
2. For all $x \in \mathbb{R}^n$, $x^\top A x \geq 0$.

NB: Similarly, A is *negative definite* if $x^\top A x < 0$ for all nonzero x.

Link with eigenvalues

Proposition

A symmetric matrix is

- positive semi-definite if and only if all of its eigenvalues are nonnegative,
- positive definite if and only if all of its eigenvalues are positive.

Remark

Positive definite matrices are invertible (since their eigenvalues are nonzero), whereas positive semi-definite matrices might not be.

Link with eigenvalues (bonus)

Proof: (sketch).
Let A be a symmetric matrix. By the spectral theorem, there exists an orthonormal basis of eigenvectors $\{v_1, \ldots, v_n\}$ with corresponding eigenvalues $\{\lambda_1, \ldots, \lambda_n\}$. For any $x \in \mathbb{R}^n$, we can write $x = \sum_{i=1}^n \alpha_i v_i$. Then,

$$x^T A x = \left(\sum_{i=1}^n \alpha_i v_i \right)^T A \left(\sum_{j=1}^n \alpha_j v_j \right) = \sum_{i=1}^n \lambda_i \alpha_i^2$$

Thus, $x^T A x \geq 0$ for all x if and only if $\lambda_i \geq 0$ for all i, and $x^T A x > 0$ for all nonzero x if and only if $\lambda_i > 0$ for all i.

A practical note

NB: if you already have a positive semi-definite matrix, it is possible to perturb its diagonal slightly to get a positive definite matrix.
With $\mathbf{A}$ a positive semi-definite and $\epsilon > 0$, $\mathbf{A} + \epsilon \mathbf{I}_n$ is positive definite, as for any $\mathbf{x} \neq 0$ we have:

$$\mathbf{x}^\top (\mathbf{A} + \epsilon \mathbf{I}_n)\mathbf{x} = \mathbf{x}^\top \mathbf{A}\mathbf{x} + \epsilon \mathbf{x}^\top \mathbf{I}_n \mathbf{x} = \underbrace{\mathbf{x}^\top \mathbf{A}\mathbf{x}}_{\geq 0} + \underbrace{\epsilon \|\mathbf{x}\|_2^2}_{>0} > 0$$

Example: the Gram Matrix

Definition (Gram Matrix)

If A is an $m \times n$ matrix, its *Gram matrix* is:

$$G = A^T A.$$

Proposition

The Gram matrix is symmetric and positive semi-definite.

Proof: exercise.

Example: the Gram Matrix

Definition (Gram Matrix)

If A is an $m \times n$ matrix, its *Gram matrix* is:

$$G = A^T A.$$

Proposition

The Gram matrix is symmetric and positive semi-definite.

Proof: exercise. We have, $G^T = (A^T A)^T = A^T (A^T)^T = A^T A = G$. Moreover, for any $\mathbf{x} \neq 0$,

$$\mathbf{x}^\top G \mathbf{x} = \mathbf{x}^\top A^\top A \mathbf{x} = (A\mathbf{x})^\top (A\mathbf{x}) = \|A\mathbf{x}\|_2^2 \geq 0$$

with equality if and only if $A\mathbf{x} = 0$, i.e. $\mathbf{x} \in \mathrm{Ker}(A)$.

Following the previous remark, the matrix $\mathbf{A}^\top \mathbf{A} + \epsilon \mathbf{I}_n$ is positive definite (and in particular, invertible) for *any* matrix $\mathbf{A}$ and any $\epsilon > 0$. It is often used in practice in data science.

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Polar decomposition (bonus)

Theorem (Polar Decomposition)

For any square matrix $A \in \mathbb{R}^{n \times n}$, there exists an orthogonal matrix R and a symmetric positive semi-definite matrix F such that

$$A = RF.$$

One can take

$$F = (A^\top A)^{1/2}.$$

NB: If A is invertible, then R is unique. If A is singular, R is generally not unique.

Reminder: repeated transformations and stability

Consider an iterative linear system

$$x_{t+1} = Ax_t.$$

If A is diagonalizable, $A = PDP^{-1}$, then

$$A^t = PD^tP^{-1}.$$

Therefore the long-term behavior is controlled by the eigenvalues:

$$D^t = \mathrm{diag}(\lambda_1^t, \ldots, \lambda_d^t).$$

- If $|\lambda_i| < 1$, the corresponding direction disappears.
- If $|\lambda_i| > 1$, the corresponding direction explodes.
- If $|\lambda_i| = 1$, the corresponding direction is preserved in magnitude.

This is a simplified linear-algebra model for stability questions in recurrent models and iterative algorithms.[1]

[1] Further reading: R. Pascanu, T. Mikolov, Y. Bengio, *On the difficulty of training Recurrent Neural Networks*, arXiv:1211.5063. https://arxiv.org/abs/1211.5063

ML illustration: Lipschitz functions and spectral norm

A function $f : \mathbb{R}^d \to \mathbb{R}^k$ is called L-**Lipschitz** if

$$\|f(x) - f(y)\|_2 \leq L\|x - y\|_2 \qquad \text{for all } x, y.$$

For a linear layer

$$x \mapsto Wx,$$

the Lipschitz constant is controlled by the spectral norm:

$$\|Wx - Wy\|_2 = \|W(x - y)\|_2 \leq \|W\|_2\|x - y\|_2.$$

ML interpretation: Controlling $\|W\|_2$ limits how much a layer can amplify perturbations. If each layer has a small spectral norm, the network is less sensitive to small input perturbations (useful for robustness). This idea is used in robust neural networks and stable GAN training.[2]

[2] Further reading: T. Miyato et al., *Spectral Normalization for Generative Adversarial Networks*, arXiv:1802.05957. https://arxiv.org/abs/1802.05957; M. Cisse et al., *Parseval Networks: Improving Robustness to Adversarial Examples*, arXiv:1704.08847. https://arxiv.org/abs/1704.08847

ML illustration: Lipschitz constant of a neural network

Consider a feedforward neural network

$$f(x) = W_L \sigma(W_{L-1} \sigma(\cdots \sigma(W_1 x))).$$

If the activation σ is 1-Lipschitz, for example ReLU, then

$$\mathrm{Lip}(f) \leq \prod_{\ell=1}^{L} \|W_\ell\|_2.$$

Interpretation: if each layer has a small spectral norm, the whole network is less sensitive to small changes in the input.

This is one motivation behind spectral normalization and Lipschitz-constrained neural networks.[3]

[3] Further reading: M. Cisse et al., *Parseval Networks: Improving Robustness to Adversarial Examples*, arXiv:1704.08847. https://arxiv.org/abs/1704.08847

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Singular Value Decomposition

Theorem (Singular Value Decomposition)

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. *Then there exist orthogonal matrices* $\mathbf{U} \in \mathbb{R}^{m \times m}$ *and* $\mathbf{V} \in \mathbb{R}^{n \times n}$, *and a diagonal matrix* $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$ *with nonnegative diagonal entries* $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(m,n)} \geq 0$ *such that*

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top.$$

The diagonal entries (σ_i) *are the* **singular values** *of* $\mathbf{A}$. *The columns of* $\mathbf{U}$ *(resp.* $\mathbf{V}$*) are the* **left** *(resp.* **right***) singular vectors of* $\mathbf{A}$.

NB: *every matrix* $\mathbf{A} \in \mathbb{R}^{m \times n}$ has an SVD (even non-square matrices)!

Shapes: $\mathbf{U} = [u_1 \ \ldots \ u_m]$, $\mathbf{V} = [v_1 \ \ldots \ v_n]$ are orthogonal, and $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$ has entries

$$\Sigma_{ii} = \sigma_i \quad \text{for } i = 1, \ldots, \min(m, n), \qquad \Sigma_{ij} = 0 \quad \text{if } i \neq j.$$

Remarks on SVD

By convention, the singular values are given in non-increasing order, i.e.

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(m,n)} \geq 0$$

Only the first r singular values are nonzero, where r is the rank of $\mathbf{A}$. We can write the rank-r part as $\mathbf{A} = \sum_{i=1}^{r} \sigma_i \, u_i v_i^\top$. Remark that the SVD factors provide eigendecompositions for $\mathbf{A}^\top\mathbf{A}$ and $\mathbf{A}\mathbf{A}^\top$:

$$\mathbf{A}^\top\mathbf{A} = (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \mathbf{V}\mathbf{\Sigma}^\top\mathbf{U}^\top\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \mathbf{V}\mathbf{\Sigma}^\top\mathbf{\Sigma}\mathbf{V}^\top$$
$$\mathbf{A}\mathbf{A}^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top\mathbf{V}\mathbf{\Sigma}^\top\mathbf{U}^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^\top\mathbf{U}^\top$$

Hence the columns of $\mathbf{V}$ (the **right-singular vectors** of $\mathbf{A}$) are eigenvectors of $\mathbf{A}^\top\mathbf{A}$, and the columns of $\mathbf{U}$ (the **left-singular vectors** of $\mathbf{A}$) are eigenvectors of $\mathbf{A}\mathbf{A}^\top$.

SVD — Geometric intuition

- Think of $\mathbf{A}$ as a three-step transform:

$$\mathbf{A} = \underbrace{\mathbf{U}}_{\text{rotate/reflect in } \mathbb{R}^m} \quad \underbrace{\mathbf{\Sigma}}_{\text{axis-wise stretch}} \quad \underbrace{\mathbf{V}^\top}_{\text{rotate/reflect in } \mathbb{R}^n} .$$

- The unit sphere in $\mathbb{R}^n$ is mapped by $\mathbf{V}^\top$ to itself (rotation/reflection), then by $\mathbf{\Sigma}$ to an axis-aligned ellipsoid (semi-axes σ_i), then by $\mathbf{U}$ to a rotated ellipsoid in $\mathbb{R}^m$.
- Right singular vectors v_i give the principal input directions, left singular vectors u_i give the principal output directions; σ_i are the semi-axis lengths (gains).
- For $k < r$, truncating after k terms keeps the k *most amplifying* directions.

SVD — Proof sketch (via spectral theorem)

(Sketch). Use that $\mathbf{A}^\top\mathbf{A}$ is symmetric positive semidefinite.

1. $\mathbf{A}^\top\mathbf{A}$ is symmetric $\Rightarrow$ by the spectral theorem, there exists an orthonormal basis $\{v_1, \dots, v_n\}$ of eigenvectors with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Set $\sigma_i = \sqrt{\lambda_i}$ and $\mathbf{V} = [v_1 \ \dots \ v_n]$.
2. Define $\mathbf{\Sigma} = \mathrm{diag}(\sigma_1, \dots, \sigma_{\min(m,n)})$ (embedded in $\mathbb{R}^{m\times n}$ by padding zeros).
3. For each i with $\sigma_i > 0$, set

$$u_i = \frac{\mathbf{A}v_i}{\sigma_i} \in \mathbb{R}^m.$$

   Then $\|u_i\|_2 = 1$ and $\langle u_i, u_j\rangle = \delta_{ij}$ for $\sigma_i, \sigma_j > 0$. Complete to an orthonormal basis $\{u_1, \dots, u_m\}$ of $\mathbb{R}^m$ and set $\mathbf{U} = [u_1 \ \dots \ u_m]$.
4. By construction, $Av_i = \sigma_i u_i$ for all i with $\sigma_i > 0$. If $\sigma_i = 0$, then

$$\|Av_i\|_2^2 = v_i^\top A^\top A v_i = 0,$$

   so $Av_i = 0$. Hence $AV = U\Sigma$, and therefore $A = U\Sigma V^\top$.

Applications of the SVD

- **Low-rank approximation.** The SVD provides the best low-rank approximations to a matrix in both the Frobenius norm and the 2-norm.
- **Pseudoinverse.** The SVD provides a way to compute the Moore-Penrose pseudoinverse of a matrix.
- **Principal component analysis (PCA).** PCA is a technique for dimensionality reduction that uses the SVD to find a low-dimensional representation of data that captures as much variance as possible.

Next class = Exam

Exam : Next Friday

- Duration = 2h.
- Starts at 1.45pm.
- But: be there at 1.30pm.

Important notes

Check the timetable! Room.
Check the course homepage for detailed instructions about the exam.
The exam is **closed-book**. But: You are allowed to bring **a single A4 sheet** with handwritten notes (recto-verso).